ZGZ RUN1: pulse times and energies with pile-up recovered

Cover note to the result tables. Start here; the report has the details.

What was done

Every record of channels 0, 1 and 2 was put in one class: a clean single pulse, a pile-up record, a record that starts on the tail of an earlier pulse, or a saturated pulse. Clean pulses give the pulse shape, and the tail and undershoot that the records cut off were measured from the baseline of the next record. With these, template matching (TM) fitted every record that is not clean and returned the time and amplitude of each pulse in it. The amplitudes were converted to energy with the 766 keV peak. Synthetic pile-up made from real clean pulses shows that a second pulse larger than half the photopeak is found in 98 % of the pairs 4–10 µs apart.

What you get

One table per channel, `events_ch<n>.csv`, with one row per pulse. The numbers:

	ch 0	ch 1	ch 2
records in `c<ch>.txt`	2,611,405	2,036,169	2,469,207
clean single pulses	2,407,593	1,792,079	2,281,258
pile-up records resolved into ≥ 2 pulses	65,760	82,554	73,874
pulses from those records	139,192	178,162	157,455
pulses from pile-up in records that start on an earlier tail	21,707	28,623	20,158
other fitted single pulses (on an earlier tail, slow rise, …)	100,215	122,876	90,705
pulses in the table (empty flag)	2,668,707	2,121,740	2,549,576
calibration (keV per ADC)	22.77	24.54	24.18
photopeak FWHM, clean pulses	6.1 %	6.0 %	5.6 %
photopeak FWHM, pulses from pile-up	6.0 %	5.9 %	5.7 %

The columns you need are `kind` (`clean`, `fitted`, `baseline`, `saturated`), `n_pulses` (pulses in that record; rows with `n_pulses` ≥ 2 are resolved pile-up, of kind `fitted`, or `baseline` when the record starts on an earlier tail), `time_s` (time since the start of the run, on the clock of `out.txt`, common to all channels), `energy_kev` and `flag` (use only rows where it is empty). The usage guide lists all columns.

```
import pandas as pd
ev = pd.read_csv("events_ch1.csv")
ev = ev[ev.flag.isna()]
clean   = ev[ev.kind == "clean"]
pileup  = ev[(ev.kind == "fitted") & (ev.n_pulses >= 2)]     # pile-up rows of the table
all_pileup = ev[ev.n_pulses >= 2]          # plus pile-up on an earlier tail
everything = ev[ev.kind != "saturated"]       # all pulses with an energy
```

What to say in a paper

Suggested text, to adapt: “Pulses that overlap in the same record (pile-up) were resolved with template matching. Templates were built from clean single pulses of each detector, completed with the pulse tail and undershoot measured from the baseline of consecutive records, and every record that is not a clean single pulse was fitted with a sum of templates, giving the time and amplitude of each pulse. Tests on synthetic pile-up built from measured pulses show that two pulses are separated from about 4 µs apart, with the photopeak resolution of clean pulses preserved.” The method and its checks are in the report; we can agree on the reference to cite.

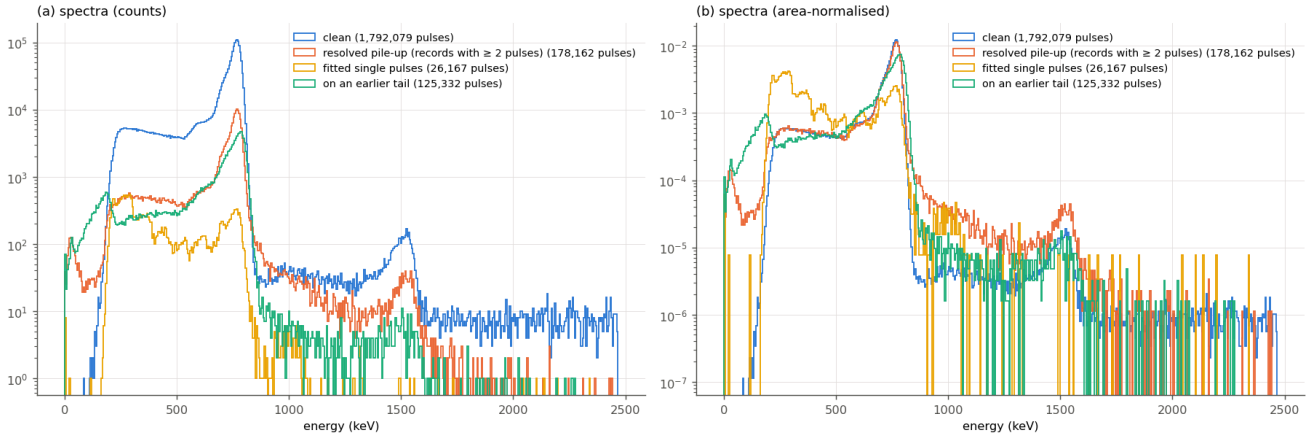
What to watch

- **Two pulse shapes.** Some clean pulses decay faster than the main shape (34 %, 3 % and 25 % in channels 0, 1 and 2, almost all below 400 keV). One template set describes both, so their energies are approximate. `relative_area` < 0.92 selects them among clean pulses.
- **Pulses closer than about 1 µs** cannot be separated and come out as one pulse with the summed energy.
- **Saturated pulses** have a time but no energy.
- **Calibration** is a single factor from the 766 keV peak; the sum peak lands about 1 % below twice that energy. Known lines can replace it.

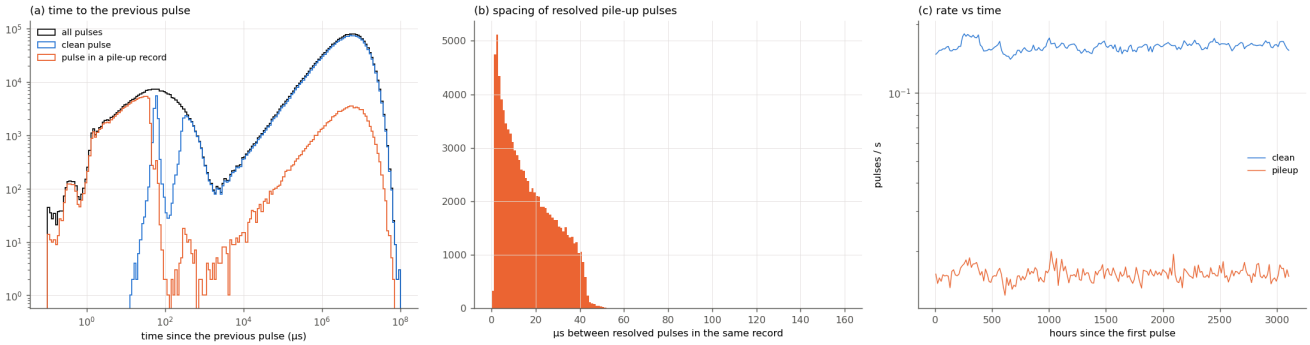
To run it on another run

The code is in the `zgz-pileup` repository: fork or clone it and install it once (usage guide, section 1), add the run to `RUNS` in `config.py`, and run the eight scripts in order, or all at once with `./run_all.sh`. Each script prints what it found and draws its figures, so you can stop after any of them and look. A run of three channels takes about 45 minutes on a desktop and gives the same table and figures.

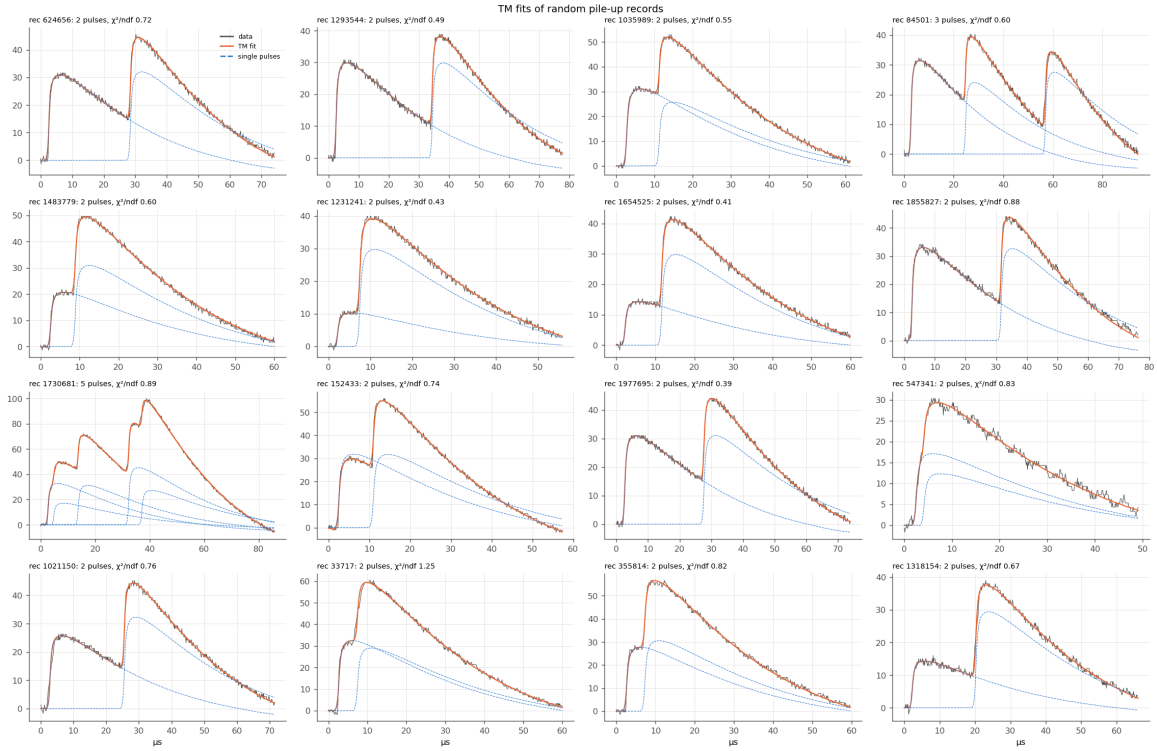
Figures you can use



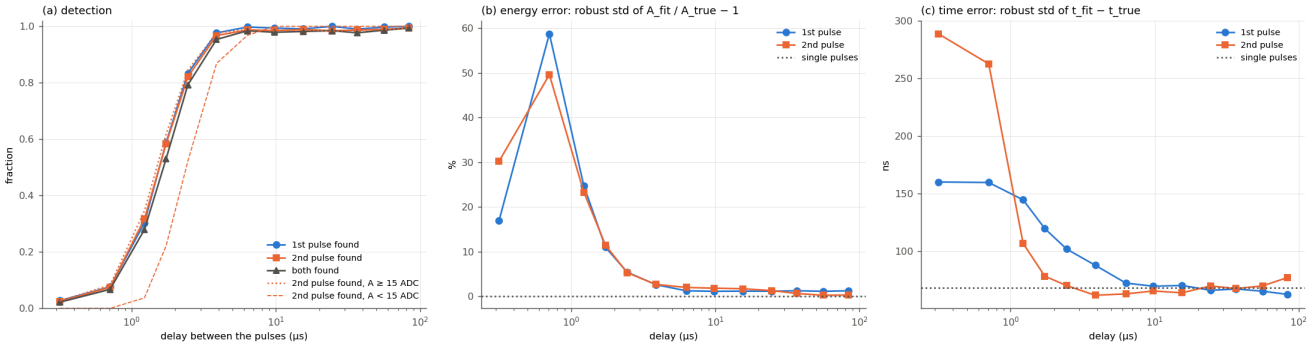
Energy spectra (channel 1). Clean pulses and pulses resolved from pile-up have the photopeak at the same place and with the same width.



Time distributions (channel 1). Times between pulses, including the short intervals that only pile-up recovery gives.



Fits of real pile-up records (channel 1), chosen at random: data, TM fit and the pulses found.



Benchmark on synthetic pile-up (channel 1): fraction of pairs resolved and energy and time errors against the delay between the two pulses.